INCH- POUND

MIL-DTL-9813C (USAF)
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SUPERSEDING
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5 June 1996

DETAILED SPECIFICATION

TARGET SYSTEM, EXTERNAL TOW, GUNNERY, A/A3-7-U-15

Inactive for new design and is no longer used, except for replacement purposes.

Beneficial comments (recommendations, additions, and deletions) any pertinent data, which may be used in improving this document, should be addressed to: 984 CSUG/GBCLE, 6040 Gum Lane, Bldg 1216, Hill AFB, UT 84056-5825 or CSUG.GBCL.Workflow@HILL.af.mil.

AMSC N/A

FSC 6920

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1. SCOPE

1.1 Scope. This specification cover the performance requirements for an externally mounted gunnery tow target system, designated A/A37U-15.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. This section does not include documents specified in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-T-9918 (Inactive)	Tow Target, Aerial Dart Gunnery Type TDU-10B

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	Environmental Engineering Consideration
MIL-STD-8591	Airborne Stores, Suspension Equipment & Aircraft-Store Interface (Carriage Phase)

(Copies of these documents are available online at http://www.assist.daps.dla.mil.online/start/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE DRAWINGS

61J10536	Target Reel, Model 29C2
61J10905	Rewind Stand, THU – 4/E

(Copies of the drawings are available from 510CBSS/GBHAA, 6020 Gum Lane, Hill AFB, UT 84056-5825.)

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2.3 Non-Government standards, specification and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.)

Anderson, Greenwood & Company

29-0402	Tow Reel, Model 29C2
29B0495	Adapter Kit, F-4 Aircraft
29D0768	Installation Kit, F-5 Aircraft

(Copies of the non-government drawings are available through the contracting officer or the procuring activity.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified, samples shall be subjected to first article inspection in accordance with paragraph 4.2.

3.2 Design. The target system shall conform to drawing, 61J10536.

3.2.1 Components. The target system shall consist of the following major components:

ITEM NO.	DESCRIPTION	QUANTITY
1	Pod, Tow Reel RMU-10/A	1
2	Boom-Target Adapter	1
3	Launcher-Target Boom	1
4	Tow Reel, Model 29C2 (Anderson-Greenwood)	1

3.2.1.1 Tow Reel Pod. The tow reel pod shall be designed to minimize aerodynamic drag in the operating environment. The design and location of the tow reel brake drum shall be such as to prevent overheating of the brake assembly.

3.2.1.1.1 Cable capacity. The capacity of the tow reel spool shall be a minimum of 2200 feet of 1-1/64 inch armored tow cable.

3.2.1.1.2 Servicing provisions. Capability shall be provided for removal of the empty spool and replacement with a pre-wound cable spool without removing the reel and reel housing from the aircraft.

3.2.1.2 Boom-target adapter. The boom-target adapter shall provide the mechanical and electrical connections between the tow reel pod and the launcher-target boom, and shall meet the requirements of drawing 61J10536.

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3.2.1.3 Launcher-target boom. The launcher-target boom shall provide the mechanical and electrical connections between the boom-target adapter and the target, and shall meet the requirements of drawings 61J10536.

3.2.2 Installation kit. Installation kit for the target system shall be adaptable, without mechanical and electrical modification, to the following aircraft.

AIRCRAFT	DRAWING NO.
F-4	29B0495
F-5	29D0768

3.3 Performance characteristics. The target system (without the tow target attached) shall conform to a 1000 pound class store in flight as specified in MIL-STD-8591. This system shall be designed to perform under a maximum aerodynamic load of 4g, at Mach 1.3, 30,000 feet mean sea level (msl).

3.3.1 Operation. The target system shall comply with the following functional requirements:

a. Provide mounting and firm support of the TDU-10/B dart tow target during taxi, take-off, and flight to 350 knots indicated air speed (KIAS) at a load of 1.5g.

b. Provide smooth release and deployment of the target to the full length of the tow cable. Release and deployment shall be accomplished at a minimum of 200 KIAS at altitudes from 1,000 to 2,000 feet msl.

c. Provide secure attachment of the TDU-10/B dart tow target during aircraft towing to a maximum airspeed of 450 KIAS at 0.9 Mach with 2.5g loading.

d. Allow initiation of target recovery by severing the tow cable at airspeeds above 200 KIAS and altitudes between 1,000 and 2,000 feet msl.

e. Provide secure attachment of the target system without the target and cable attached during aircraft flight speeds to Mach 1.3, at loads of 4g, and at a altitudes to 30,000 feet msl.

3.3.2 Environmental conditions. The target system shall meet the requirements of this specification under the following conditions:

a. Any altitudes between zero and 30,000 feet msl.

b. Vibration incident to the target system installed on the aircraft listed in paragraph 3.2.2.

3.3.3 Acceleration loads. The target system shall withstand acceleration loads typical of normal aircraft installations for a 1,000 pound class store in flight, as specified in MIL-STD-8591.

3.4 Supportability. All part having the same manufacturer’s part number shall be functionally and dimensionally interchangeable.

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3.4.1 Accessibility. The tow target system shall afford accessibility to installed components to facilitate ease of maintenance.

3.4.2 Identification. Equipment, assemblies, containers, and parts shall be marked for identification. The nomenclature and stock number to be included on the equipment nameplate shall be as follows:

Target System, External Tow, Gunnery A/A37U-15
Stock Number 6920-00-056-6036

3.5 Color and finish. Color and finish shall be in accordance with applicable drawings.

3.6 Safety. The tow target system shall be functional without sharp edges, burrs, or protrusions that might injure personnel.

3.7 Workmanship. The target system, including all parts and equipment, shall be constructed in accordance with commonly accepted industrial workmanship standards.

3.8 Government-furnished property. When specified in the contract or purchase order, the Government will furnish the following to the contractor upon request, for use in fabrication of the two target system:

STOCK NO.	NOMENCLATURE
1095-00-141-9328	Rack Assembly, Bomb, Type MA-4A

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection, see paragraph 4.2.

b. Conformance inspection, see paragraph 4.3.

4.2 First article inspection.

4.2.1 First article test samples. First article test samples shall consist of the first five target systems representative of the production equipment. Three target systems shall be used for environmental and static load tests by the contractor, and two target systems shall be delivered for inspection and flight testing by the Government, as directed by the procuring agency.

4.2.2. First article inspection requirements. First article inspection shall consist of all examination and tests described in paragraph 4.4.

4.3 Conformance inspection. Each target system shall be subjected to the individual examination and tests as described in paragraphs 4.4.1 and 4.4.2.

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4.4 Methods of inspection.

4.4.1 Examination. Each target system shall be examined for compliance with the requirements specified in paragraphs 3.2, 3.3, 3.4, 3.5, and 3.6. This element of inspection shall encompass all visual examinations and dimensional measurements. Non-compliance with any specified requirement, or presence of one or more defects preventing or lessening maximum efficiency, shall constitute cause for rejection.

4.4.2 Ground operation. Power shall be applied to the target system to determine compliance with drawing 61J10536 bomb rack and cable cutter circuitry.

4.4.3 Environmental tests. All environmental tests will be accomplished with an empty tow reel spool. Failure of any of these tests will be cause for rejection.

4.4.3.1 Vibration tests. The target system shall be mounted on the vibration table in the same manner as mounted on the applicable aircraft. The vibration procedure shall be as follows:

a. Vibrate the system in a horizontal direction with the frequency varying between 10 and 55 cycles per second (cps) at single amplitude of 0.03 inch (0.06 inch total excursion). The frequency shall be changed uniformly from 10 to 55 cycles and returned to 10 cycles in 1 to 3 minutes. This test shall continue for 90 minutes. During this time the frequency of any and all resonant point (natural periods) shall be noted. If more than three resonant points are encountered with vibration applied along one axis, the three most severe shall be chosen for the test.

b. Vibrate the equipment for 15 minutes at each of resonant frequencies noted in step (a) at 0.06 inch total excursion.

c. Repeat steps (a) and (b) changing the direction of vibration 90° horizontally.

d. Repeat steps (a) and (b) changing the direction of vibration to vertical.

e. Remove the equipment from the table and visually inspect for any mechanical failures. There shall be no visible sign of cracking, separation, or other physical failures.

4.4.4 Static load tests. The target system shall be mounted to a structural frame in the same manner as mounted on the applicable aircraft. Critical loading conditions as specified in MIL-STD-8591 will be applied and deflection will be recorded. There shall be no permanent set, deformation, or structural damage.

4.4.5 Flight tests.

4.4.5.1 Without target test. The target system without target will be flown on the applicable aircraft at speeds to Mach 1.3, at loads to 4g, and at altitudes to 30,000 feet msl. The tow reel will have a full load of cable.

4.4.5.2 Target support test. A TDU-10/B dart tow target will be attached to the A/A37U-15 target tow system and the applicable aircraft will taxi, take-off, and fly to 350 KIAS at loads to 1.5g.

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4.4.5.3 Target launch test. The TDU-10/B dart tow target conforming to MIL-T-9918 will be launched from the target system at a minimum of 200 KIAS at an altitude between 1,000 to 20,000 feet msl.

4.4.5.4 Tow target test. The target system will be flown with the target in the tow position to speeds of 450 KIAS at 0.9 Mach at loads of 2.5g.

4.4.5.5 Target recovery test. The target system will be used to initiate target recovery at a minimum of 200 KIAS and at an altitude between 1,000 and 20,000 feet msl.

4.4.5.6 Post-flight inspection. After completion of flight testing, the target system shall be examined and tested in accordance with paragraph 4.4.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. Actual packaging activity will ascertain requisite packaging requirements; packaging requirements are maintained by the Inventory Control Point packaging activity with the Military Department or Defense Agency. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The target tow system covered by this specification is intended to provide a means to tow, with F-4 or F-5 aircraft, a TDU-10/B Dart Gunnery Tow Target for use in gunnery training or combat crew members.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of specification, standards, and handbooks to be cited in the solicitation, if required specified issues of documents referenced in paragraphs 2.2.

c. Packaging requirements see paragraph 5.1.

d. When first article inspection is required, see paragraph 3.1.

6.3 Ground support equipment. A THU-4/E Re wind Stand, AF drawing 61J10905, is used in conjunction with this system as ground support equipment.

6.4 Government furnished property. The contracting officer should arrange to furnish the property listed in paragraph 3.7.

6.5 Subject term (key word) listing.

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Aircraft
Gunnery
Tow Target

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIALS

Custodian:
Air Force – 70

Preparing Activity:
Air Force – 70

Project No. 6920-2008-001

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, verification of the currency of the information above using the ASSIST Online database at: http://assist.daps.dla.mil.

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